

MAP OPERATIONS SPECIALIST / ENGINEER

POI Data Quality - Address Intelligence - Search Relevance - Geofencing

JOB OVERVIEW

Position	MAP OPERATIONS SPECIALIST
Focus	POI Data Quality, Geocoding, Search Relevance, Geofence Governance
Department	TASCO.COM.VN MAPPING
Location	TASCO Building Hanoi, Pham Hung
Level	Individual Contributor
Reports to	Head of Mapping

1. JOB PURPOSE

The POI / Geocoding / Search Operations Specialist owns the quality, completeness, and usability of the place and address data that powers Tasco Maps, MyTasco, VETC mobility services, local search, discovery, routing destinations, and vehicle-related commerce.

This role is responsible for building and maintaining high-quality datasets for points of interest, addresses, entrances, access points, administrative hierarchy, search ranking signals, and geofences. The role works closely with Product, Engineering, Data Science, MapOps, and Business teams to ensure users can accurately search, discover, navigate to, and transact with real-world locations across Vietnam.

The role is not only about maintaining POIs. It is about building a Vietnam-localized place intelligence system for maps, navigation, parking, fuel, EV charging, toll stations, vehicle services, insurance, logistics, and local commerce.

2. KEY RESPONSIBILITIES

A. POI Data Quality Governance

- Own and define POI data-quality standards for Tasco Maps across priority regions: Hanoi, Ho Chi Minh City, expressways, national highways, VETC/Tasco locations, parking sites, fuel stations, EV charging stations, car service centers, toll stations, offices, restaurants, cafes, and commercial places.
- Design and maintain POI classification taxonomy, naming standards, category hierarchy, address format, opening hours, phone number, website, payment method, brand, chain ID, and operational status.
- Define the minimum data requirement for each POI category, including display name, category, address, latitude/longitude, entrance/access point, road-side access direction, business status, source, freshness, and validation status.
- Lead large-scale POI cleaning, deduplication, merging, normalization, and enrichment workflows.
- Monitor POI completeness, duplication rate, wrong-location rate, stale data rate, and category accuracy by city, district, and business category.
- Build SOPs for POI creation, update, merge, closure, relocation, and deletion.

B. Geocoding & Address Intelligence

- Own the quality of forward and reverse geocoding datasets for Vietnamese addresses.
- Build address normalization rules for Vietnam-specific address structures: province/city, district where still relevant historically, ward/commune, street, alley/lane/ngo/ngach/hem, building, house number, apartment, floor, and unit.
- Define and maintain administrative hierarchy datasets after administrative boundary changes.
- Improve address-to-coordinate accuracy for residential addresses, business addresses, VETC/Tasco service points, logistics pickup/dropoff, parking entrances, toll station entrances, and fuel/charging station entrances.
- Work with Engineering to improve geocoding services, including structured input, full-text input, typo tolerance, address parsing, reverse lookup, batch geocoding, and entrance/routable point selection.
- Establish accuracy metrics for geocoding: rooftop-level accuracy, street-level accuracy, ward-level accuracy, wrong-side-of-road rate, entrance mismatch rate, and failed geocode rate.

C. POI Search & Discovery Quality

- Own search quality for POI and place discovery in Tasco Maps, MyTasco, VETC, and related mobility applications.
- Define relevance metrics for search results: top-1 accuracy, top-3 accuracy, click-through rate, zero-result rate, wrong-result rate, category precision, distance-to-intent, and reroute-after-search rate.
- Work with Search/AI Engineering to improve autocomplete, fuzzy matching, synonym handling, Vietnamese accent-insensitive search, abbreviation handling, brand search, category search, nearby search, search ranking, and personalized or context-aware search.

- Maintain query dictionaries and search relevance rules for Vietnamese mobility contexts such as parking, EV charging, fuel stations, car wash, toll station, vehicle inspection, garage, roadside assistance, insurance, VETC, and Tasco.
- Partner with Product and Design to improve search UX, search filters, POI cards, category landing pages, and discovery flows.
- Analyze search logs and user feedback to identify missing POIs, wrong categories, poor ranking, and failed user intent.

D. POI Geofence & Polygon Governance

- Define and maintain geofences for operational and commercial use cases: parking zones, toll station zones, service coverage areas, dealership/service-center catchment areas, restricted service zones, logistics pickup/dropoff zones, airport/train station/mall polygons, and high-demand mobility zones.
- Own the spatial quality of POI polygons and geofences: geometry validity, polygon overlap, polygon containment, entrance/exit points, road access, service-area boundaries, and point-in-polygon accuracy.
- Build governance rules for geofence lifecycle: creation, update, approval, versioning, expiry, and rollback.
- Work with Engineering to expose geofence data for map rendering, search filtering, notification targeting, parking/fuel/charging discovery, toll-related services, campaign targeting, and fleet/logistics operations.
- Define geofence KPIs: false inclusion rate, false exclusion rate, polygon freshness, business-rule compliance, and geofence-trigger accuracy.

E. Data Sources, Conflation & Validation

- Integrate and validate data from OpenStreetMap, Overture Maps, government and administrative datasets, Tasco/VETC internal business data, partner data, field collection, street imagery, user contributions, business-owner submissions, and transaction or visit signals where allowed.
- Design POI conflation rules to merge duplicates from different sources.
- Establish evidence requirements for POI validation: image evidence, source URL, field verification, contributor confirmation, business-owner claim, transaction/visit signal, and partner contract data.
- Build source reliability scoring for POI and geocoding data.
- Partner with Data Science to develop confidence scores for POI, address, and geofence objects.
- Maintain data lineage for every critical POI and geofence object.

F. Troubleshooting, RCA & Product Feedback

- Investigate user and internal reports related to wrong POI location, missing POI, duplicate POI, incorrect address, wrong category, wrong entrance, bad search result, failed reverse geocode, and incorrect geofence trigger.
- Conduct root-cause analysis for high-impact POI/search/geocoding incidents.
- Translate field/user feedback into data updates, search-ranking improvements, or product requirements.
- Work with Engineering to identify whether issues come from data, index, ranking, query parsing, UI, or API behavior.
- Define escalation workflows for P0/P1 location-search issues.

G. Documentation, SOPs & Team Enablement

- Build and maintain documentation for POI data standard, category taxonomy, address standard, geocoding QA, search quality QA, geofence data standard, contributor validation, and POI release checklist.
- Train junior MapOps members and contributors on POI creation, validation, and QA standards.
- Review operator output and raise team quality standards.
- Work with Product, Engineering, Data Science, and Business teams to shape the POI/Search roadmap.

3. REQUIREMENTS

Must-have

- Minimum 2-4 years of hands-on experience in POI operations, geocoding, local search, GIS data operations, map operations, ride-hailing, logistics, delivery, navigation, or location-based products.
- Strong understanding of POI data structures, address data, administrative boundaries, and spatial data quality.
- Experience with QGIS, PostgreSQL/PostGIS, spreadsheets, and spatial QA workflows.
- Working knowledge of OpenStreetMap data, POI tags, address tags, administrative boundaries, and Overpass API.
- Ability to define data-quality metrics and operate large-scale data-cleaning workflows.
- Strong understanding of search-quality concepts: autocomplete, fuzzy search, category search, ranking, zero-result analysis, and click-through analysis.
- Experience with POI deduplication, conflation, or data normalization.
- Ability to read technical API documentation in English.
- Strong communication skills with Product, Engineering, Operations, and Business teams.

Nice-to-have

- Experience with Elasticsearch, OpenSearch, Pelias, Nominatim, Photon, or similar search/geocoding systems.
- Experience with Mapbox Search Box API, Geocoding API, Google Places API, HERE Geocoding, TomTom Search API, GrabMaps Places, or equivalent APIs.
- Experience with geofence creation, polygon QA, point-in-polygon analysis, or service-area modeling.
- Experience with Python for ETL, POI cleaning, deduplication, and spatial analysis.
- Experience with user-generated content, contributor workflows, or business-owner verification.
- Familiarity with Vietnamese address normalization and administrative boundary changes.
- Familiarity with mobility, parking, tolling, EV charging, fuel, logistics, insurance, or automotive service data.
- OSM contribution history or open-source mapping experience.

Soft Skills & Mindset

- Detail-oriented and skeptical about data quality.
- Strong ownership of data correctness and user experience.
- Able to balance accuracy, speed, and operational cost.
- Comfortable working with ambiguous, messy, incomplete real-world data.
- Data-driven mindset; decisions should be grounded in metrics and evidence.
- Able to translate business needs into data standards and operational workflows.
- Able to work independently and lead small data-quality initiatives.

4. SUCCESS METRICS - 6 TO 12 MONTHS

- Improve POI completeness in priority categories and regions.
- Reduce duplicate POI rate.
- Reduce wrong-location POI incidents.
- Improve top-1 and top-3 search result accuracy.
- Reduce zero-result rate for common Vietnamese mobility queries.
- Improve reverse-geocoding accuracy for pickup/dropoff and destination use cases.
- Build and roll out POI category taxonomy and address standard.
- Build geofence governance process for parking, toll, fuel, EV charging, service zones, and commercial zones.
- Establish POI/Search QA dashboard.
- Deliver SOPs for POI creation, validation, merge, deletion, geocoding QA, and geofence QA.
- Support reliable POI/search/geocoding release workflow for Tasco Maps and MyTasco.

5. PRODUCT SCOPE

Product Line	Responsibility
V-Base-Maps	POI display, categories, place cards, administrative hierarchy
V-Navigation-Maps	Destination quality, entrances, routable points, reverse geocode
Location & AI Search	Search text, autocomplete, category search, ranking
Place & Discovery	Office, restaurant, cafe, parking, fuel, EV charging, car wash, toll station
Contributor Platform	User-submitted POI, validation workflow, business-owner claim
Commercial Mobility Layer	Geofences for parking, tolling, vehicle services, campaigns, logistics

6. TECHNICAL WORKING AREAS

- POI database
- Administrative boundary database
- Address normalization rules
- Elasticsearch/OpenSearch search index
- Geocoding/reverse-geocoding service
- POI conflation pipeline
- Geofence/polygon database
- Contributor validation system
- Map Editor / internal data approval tool
- QA dashboards
- Release management process

7. DIFFERENCE FROM ROUTING & NAVIGATION ROLE

Routing & Navigation JD	POI / Geocoding / Search JD
Road topology	POI/address/location object quality
Traffic signs and restrictions	POI taxonomy and address hierarchy
Routing engine tuning	Search ranking and geocoding quality
Speed/camera/turn alerts	POI discovery and destination accuracy
Map-matching vehicle movement	Address/POI matching and entrance selection
Road-rule compliance	Geofence and service-area correctness

Routing & Navigation JD	POI / Geocoding / Search JD
Route incident RCA	Search/geocode/POI incident RCA

Internal positioning: Routing roles ensure vehicles go on the correct road. POI/Search roles ensure users find the correct place, select the correct destination, enter through the correct entrance, and the system understands the correct service area.

8. HOW TO APPLY

Please send your email with title [Tasco Mapping Application - POI Search] and CV, together with a portfolio or descriptions of POI, geocoding, search, geofence, GIS, logistics, ride-hailing, delivery, or location-data projects you have worked on to: **phung.dm@tasco.com.vn** and tuyendung@vtii.vn OSM contribution links, GitHub repositories, POI/search dashboards, or sample data-quality projects are encouraged.